



## Session nine: Data Visualization with PyGWalker in python

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### What is PyGWalker?

[PyGWalker] (short for "Python Graphic Walker") is an open-source Python library that turns your pandas DataFrames into an interactive visual exploration tool, like Tableau or Power BI. But inside Jupyter Notebooks

The goal is to make EDA faster and more accessible, especially for those who may not be proficient in writing complex visualization code.

### Key Features:

- 1. Interactive Interface:** Drag-and-drop functionality for creating charts and exploring data.
- 2. Seamless Integration:** Works directly with Pandas DataFrames in Jupyter Notebook or JupyterLab.
- 3. Wide Range of Visualizations:** Supports bar charts, line charts, scatter plots, heatmaps, and more.
- 4. Customizable:** Allows fine-tuning of visualizations and export options.
- 5. Lightweight:** Minimal setup required; simply install and start using it.

### Notes

PyGWalker is particularly useful for quick insights during the initial stages of a project.

It works best with clean datasets, so make sure to preprocess your data (e.g., handle missing values, outliers, etc) before using PyGWalker.

```
In [3]: # prompt: pygwalker  
  
        #!pip install pygwalker  
  
import pandas as pd  
import pygwalker as pyg
```

```
In [4]: #!pip install runcell
```

```
In [5]: # Load your dataset
df1 = pd.read_csv("GSSsubset.csv")
df2 = pd.read_csv("SMdata.csv")

# Launches the interactive GUI in your notebook
#pyg.walk(df)

# Initialize Pygwalker
walker = pyg.walk(df1)
```

Box(children=(HTML(value='\n<div id="ifr-pyg-000655b599cce673nhiJGtQF0NE6zC1r" style="height: auto">\n <hea...

## This will launch an interactive UI where you can:

- Drag columns to create visualizations.
- Change chart types (e.g., bar, line, scatter).
- Filter and sort data dynamically.
- Export visualizations or insights.

## Why Use PyGWalker?

1. Ease of Use : Reduces the need for writing boilerplate code for visualizations.
2. Speed : Quickly iterate through different views of your data.
3. Accessibility : Makes EDA approachable for beginners while still being powerful for advanced users.